

Chaewon Lee

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RESEARCH SUMMARY

My research develops advanced statistical and computational methods designed for the complex demands of psychological, behavioral, neural, and biomedical data. I draw on five main methodological areas, including machine learning, time-series modeling, dynamic network methods, structural equation modeling, and fuzzy statistics. I bring these strands together to create tools that strengthen traditional approaches, support reliable inference from sparse or heterogeneous datasets, and treat measurement imprecision as meaningful information rather than a nuisance. The broader aim of this work is to advance precision science across brain, behavioral, and medical research through individualized brain mapping, biomarker discovery, adaptive interventions and treatments, and prediction of clinical trajectories.

EDUCATION

Ph.D. in Quantitative Psychology	University of North Carolina, Chapel Hill, US	2026
M.A. in Quantitative Psychology		2023
▪ Advisor: Kathleen Gates ▪ Dissertation: Adaptive structural equation modeling for idiographic dynamic processes via meta-learning ▪ Thesis: An integrative machine learning approach for small samples and high-dimensional imbalanced data in psychological experiment		
M.S. in Applied Statistics	University of Michigan, Ann Arbor, US	2020
▪ Advisor: Robert Keener		
M.A. in Economics	Yonsei University, Seoul, Korea	2017
▪ Advisor: Hak Bae Lee ▪ Thesis: Impacts of Intellectual Property Rights on the Expansion of Korean Pharmaceutical Industry		
B.S. in Biology	Yonsei University, Seoul, Korea	2009
▪ Advisor: Woo Taek Kim		

RESEARCH APPOINTMENTS

Postdoctoral Research Fellow	Thyroid + Dementia Research	Jul 2026 – present
▪ Johns Hopkins University School of Medicine (PI: Jennifer Mammen) ▪ Develop and apply statistical and machine learning methods to investigate the association between thyroid hormone overtreatment and dementia risk		
Research Collaborator	Tinnitus Disorder Research	Oct 2023 – present
▪ Department of Psychology and Neuroscience, UNC Chapel Hill (PI: Thomas Rodebaugh) ▪ Conducted dynamic network modeling analyses using group iterative multiple model estimation (GIMME) to examine dynamic symptom relationships in tinnitus disorders, serving as second author on the publication		
Research Collaborator	Fuzzy Chaos Modeling Research	Jun 2024 – present
▪ Department of Mathematics and Statistics, Sejong University (PI: Jin Hee Yoon) ▪ Applied fuzzy statistics and chaos theory to identify chaotic dynamics in social opinion formation, serving as co-first author on a peer-reviewed publication and conference proceedings		

Research Assistant	NSF Student Success Project	May 2021 – Jun 2022
- School of Education, UNC Chapel Hill (PI: Jeffrey Greene, Co-PI: Kathleen Gates) - Developed sequence-based Markov machine learning approach (seqMAC), a predictive analytics framework that forecasts outcomes in online education using Markov modeling, machine learning, and explainable AI		
Graduate Researcher	Suicide Prediction Project	Apr 2019 – May 2020
- Department of Psychology, University of Michigan (PI: Patricia Deldin) - Analyzed event-related potentials to classify psychiatric patients and predict suicidality in individuals with bipolar disorder using supervised machine learning		
Statistician	PROACTIVE Data Project	Oct 2019 – Mar 2020
- Harvard Commonwealth Research Center, Beth Israel Deaconess Medical Center (PI: Theo Manschreck) - In collaboration with clinical researchers, analyzed metabolic profiles related to schizophrenia treatment effectiveness (injectable risperidone vs. oral second-generation antipsychotics).		

PUBLICATIONS

Yoon, J.H.*, **Lee, C.***, Kwon, S., & Bae, Y.C. (co-first author, Accepted). Uncertainty-Aware Nonlinear and Chaotic Dynamics in Human-Centric Opinion Systems. *International Journal of Fuzzy Systems*.

Lee, C.* & Gates, K.M. (2026). Group-iterative multiple model estimation approaches in clinical science. *Annual Review of Clinical Psychology*, Vol. 22, [\[DOI\]](#)

Lee, C.* & Gates, K.M. (2025). Automated machine learning for classification and regression: A tutorial for psychologists. *Behavior Research Methods*, 57, 262. [\[DOI\]](#)

Lee, C.*, Luo, L., Kuhlmann, S. L., Plumley, R. D., Panter, A. T., Bernacki, M. L., Greene, J. A., & Gates, K. M. (2025). Interpretable predictive analytics for online learning: A Markov-based machine learning approach. *Journal of Learning Analytics*, 12(2), 259-278. [\[DOI\]](#)

Lee, C.*, Gates, K. M., Chun, J., Al Kontar, R., Kamali, M., McInnis, M. G., & Deldin, P. (2025). Suicide risk estimation in bipolar disorder using N200 and P300 event-related potentials and machine learning: A pilot study. *Journal of Affective Disorders Reports*, 20, 100875. [\[DOI\]](#)

Plumley, R.D.*, Bernacki, M.L., Greene, J.A., ..., **Lee, C.**, Panter, A.T., and Gates, K.M. (2024). Co-designing enduring learning analytics prediction and support tools in undergraduate biology courses. *British Journal of Educational Technology*, 55(5), pp.1860-1883. [\[DOI\]](#)

SOFTWARE PACKAGE

Lee, C.* (2025). *flex* [R package]: Fuzzy least squares estimation with explicit formula (Version 0.1.0). Comprehensive R Archive Network (CRAN). [\[DOI\]](#)

MANUSCRIPTS IN REVISION / UNDER REVIEW

Rodebaugh, T.L.*, **Lee, C.**, Gates, K.M., Frumkin, M.R., ... , & Piccirillo, J. (minor revision). Modeling psychological processes in Tinnitus disorder. *Methodological Innovations*.

MANUSCRIPTS IN PREPARATION

Lee, C.*, Fisher, Z., Gates, K.M. (pending submission). Early stopping for structural equation models (target journal: Psychological Methods)

Lee, C.* & Yoon, J.H. (pending submission) Modeling measurement imprecision with least squares estimation using fuzzy numbers (target journal: *Psychometrika*)

Lee, C.* et al. (pending submission). Adaptive structural equation modeling for idiographic dynamic processes via meta-learning (target journal: *Psychometrika*)

Rodebaugh, T.*, **Lee, C.**, et al. (pending submission). Evaluating novel predictors of response of Tinnitus distress to cognitive behavioral therapy. (target journal: *JAMA Otolaryngology*)

Lee, C.* et al. (in preparation). From isolated estimation to shared learning: Multitask learning under individual heterogeneity (target journal: *Psychological Methods*)

Lee, C.* & Yoon, J.H. (in preparation). A fuzzy least squares approach to mediation and moderation. (target journal: *Psychometrika*)

Lee, C.* (in preparation). FLEX: An R package for least squares estimation with fuzzy numbers. (target journal: *Behavior Research Methods*)

PEER-REVIEWED ABSTRACTS

Lee, C.* & Gates, K. M. *Classic or Computational Graph? A Comparison of SEM Estimation Frameworks. Multivariate Behavioral Research.* [[DOI](#)]

Yoon, J. H.*, **Lee, C.***, Kwon, S., & Bae, Y. C. (co-first author, November 2025). *Nonlinear and chaotic dynamics in generalized fuzzy opinion model.* Conference paper presented at the *International Conference on Fuzzy Theory and Its Applications (iFUZZY)*, Hsinchu City, Taiwan.

Lee, C.*, Yoon, J. H., & Bae, Y. C. (November 2024). *Chaotic behaviors in opinion dynamics models influenced by external forces.* Abstract published in the *Proceedings of the 25th International Symposium on Advanced Intelligent Systems (ISIS)*, Himeji, Japan.

TEACHING EXPERIENCE

PSYC 210: Statistical Principles of Psychological Research

UNC Chapel Hill

Instructional Assistant

May 2025 – present, May 2022 – May 2024

Teaching Assistant

Aug 2020 – Apr 2021

- Supported instruction for an undergraduate Introduction to Statistics course across multiple semesters.
- Led weekly recitation sessions, provided hands-on guidance for data analysis in SPSS and Jamovi, and offered individualized guidance to help students develop statistical reasoning

RESEARCH TALKS

International Meeting of the Psychometric Society (IMPS) Annual Meeting (Jul 20-24, 2026)

Yonsei University, Seoul, South Korea

- **Lee, C.*** & Gates, K.M. “Adaptive structural equation modeling for idiographic dynamic processes via meta-learning”

International Conference on Fuzzy Theory and Its Applications (iFuzzy) (Nov 5-8, 2025)

Hsinchu, Taiwan

- Co-first author, “Nonlinear and chaotic dynamics in generalized fuzzy opinion model”
- Yoon, J.H.*, **Lee, C.***, Kwon, S., & Bae, Y.C. (presented by Yoon, J.H.)

Society of Multivariate Experimental Psychology (SMEP) Annual Conference (Oct 22-25, 2025)

University of Notre Dame, Indiana, US

- **Lee, C.*** & Gates, K.M. “No single approach fits all: Testing two generations of structural equation modeling estimation”

International Symposium on Advanced Intelligent Systems (ISIS) (Nov 9-12, 2024)

Himeji, Japan

- First author, “Chaotic behaviors in opinion dynamics models influenced by external forces”
- **Lee, C.***, Yoon, J.H., & Bae, Y.C. (presented by Yoon, J.H.)

American Educational Research Association (AERA) Annual Meeting (Apr 11-14, 2024)

Philadelphia, Pennsylvania, US

- Co-author, “Improving the Prediction of Undergraduate STEM Outcomes via Curricular Temporality-Based Feature Design” Plumley, R.D.*, Bernacki, M.L., ..., **Lee, C.**, ..., & Gates, K.M. (presented by Plumley, R.D.)

Quant Forum

L. L. Thurstone Psychometric Laboratory, UNC Chapel Hill

- “Rethinking SEM under limited data: From computational graph SEM to adaptive SEM via meta-learning” (Apr 2026)
- “Integrative classification framework in machine learning for small samples and high-dimensional imbalanced data” (April 2023)
- “Sequence analysis for classification: A hybrid Markov–machine learning approach” (April 2022)
- “Predicting suicide attempt in bipolar disorder with machine learning” (May 2021)

AWARDS AND HONORS

Dissertation Research Grant	Society of Multivariate Experimental Psychology (SMEP)	2026
▪ Awarded \$4,000 in doctoral dissertation research funding		
Graduate Student Travel Award	Society of Multivariate Experimental Psychology (SMEP)	2025
▪ Supported travel to present a graduate student poster at the Annual SMEP Conference		
Departmental Undergraduate Teaching Commendation	UNC Chapel Hill	Dec 2020, May 2021
▪ Awarded twice for excellent teaching in PSYC 210: Statistical Principles of Psychological Research		
Graduate Funding Package	Graduate School, UNC Chapel Hill	2020-present
▪ Full tuition and stipend support for PhD training via teaching and research assistantships		
Merit-Based Scholarship	Graduate School of Economics, Yonsei university	Mar 2013
▪ Awarded for outstanding academic performance.		
National Science and Engineering Scholarship	Korea Student Aid Foundation	2003-2009
▪ Awarded full undergraduate tuition and stipend for top national performance on the Korean SAT.		

COURSEWORK

M.A. & Ph.D. in Quantitative Psychology

Machine Learning; Test Theory; Multilevel Modeling;
Factor Analysis; Structural Equation Modeling;
Intra-Individual Analysis; Computational Biology;
Attention; Biomedical Imaging Science;

M.A. in Economics

Econometrics; Statistics; Game Theory;
Mathematical Statistics; Real Analysis & Measure theory;
Linear Algebra; Economic Research (SPSS);
Macroeconomics; Microeconomics; Copyright Law

M.S. in Applied Statistics

Probability Theory; Statistical Inference;
Regression Analysis; Financial Statistics;
Machine Learning; Data Analysis in Python;
Design of Experiment; Database App Design;

B.S. in Biology

Cell Biology; Genetics; Cell Physiology;
Molecular Biology; Animal Histology & Physiology;
Embryology; Microbiology; Virology
Plant Molecular Physiology; Biochemistry

PROFESSIONAL SERVICE

Ad Hoc Reviewer

Journals Behavior Research Methods; Data Mining and Knowledge Discovery; Cluster Computing; Research Methods in Applied Linguistics; BMC Psychiatry; Scientific Reports; Artificial Intelligence Review

Textbook Deep learning textbook proposal (Bentham Science Publishers)

Departmental Service

- Faculty search committee member, Quantitative Psychology, UNC Chapel Hill (2025-2026)
- Ph.D. program recruitment meetings with prospective students (2022-2024)

Conference Service

- Local organizing committee assistant, 2026 International Meeting of the Psychometric Society (IMPS)
- Poster proposal reviewer, 2026 American Psychological Association (APA)

Mentoring

- Research mentor to undergraduate students majoring in psychology and statistics (UNC Chapel Hill)

NIH R21 Grant Proposal

- Contributed to a grant proposal led by Dr. Kathleen Gates by drafting methods sections and editing.

PROFESSIONAL TRAINING AND CERTIFICATES

Functional MRI Visiting Fellowship

Oct 26-30, 2019

Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts, US

- Training in fMRI research: magnetic resonance physics, hemodynamic responses, statistical analysis, stimulus presentation and response recording, and the design of perceptual and cognitive experiments.

Certified Research Analyst

2010

- Awarded by the Korea Financial Investment Association for expertise in financial analysis, investment research, and valuation.

Certified Securities Investment Consultant

2009

- Awarded by the Korea Financial Investment Association for qualification to advise investors on securities investment and portfolio management.

PROFESSIONAL EXPERIENCE

Economist

Korea Investment & Securities, Seoul, Korea Dec 2010 – Aug 2013

- Research Center
- Published more than 100 economic reports on macroeconomic and financial markets; projected key indicators such as GDP, consumer prices, crude oil prices, exchange rates, and policy rates.
- Representative economic articles include:
 - “European Economy: Leading economic index to foretell positive cycle” (May 23, 2013)
 - “Fluctuations in raw material prices by the uncertainty of US presidential election” (Nov 7, 2012)

Investment Banker

Korea Investment & Securities, Seoul, Korea Dec 2009 – Dec 2010

- Department of Equity Capital Markets
- Managed initial public offerings on Korean stock indices (KOSPI and KOSDAQ), including financial due diligence, accounting analysis, and listing documentation.